First report on the molecular characteristics of mcr-1 colistin resistant E. coli isolated from retail broiler meat in Bangladesh

Article *in* International Journal of Food Microbiology · December 2022

DOI: 10.1016/j.ijfoodmicro.2022.110065

6 authors, including:

Wohab Ali
Bangladesh Agricultural University

Kishor Sosmith Utsho
Bangladesh Agricultural University

Susmita Karmakar
Bangladesh Agricultural University

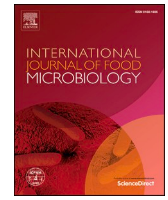
Md.Tanvir Rahman
Bangladesh Agricultural University

International Journal of Food Microbiology 388 (2023) 110065

Contents lists available at ScienceDirect

International Journal of Food Microbiology

journal homepage: www.elsevier.com/locate/ijfoodmicro



First report on the molecular characteristics of *mcr-1* colistin resistant *E. coli* isolated from retail broiler meat in Bangladesh

Md. Wohab Ali [a], Kishor Sosmith Utsho [a], Susmita Karmakar [a], Md. Nazmul Hoque [b], Md. Tanvir Rahman [a], Jayedul Hassan [a,*]

[a] *Department of Microbiology and Hygiene, Faculty of Veterinary Science, Bangladesh Agricultural University, Mymensingh 2202, Bangladesh*
[b] *Department of Gynecology, Obstetrics & Reproductive Health, Faculty of Veterinary Medicine & Animal Science, Bangabandhu Sheikh Mujibur Rahman Agricultural University, Gazipur 1706, Bangladesh*

A R T I C L E I N F O

Keywords:
E. coli
Colistin resistance
mcr-1
Plasmid
MDR
Bangladesh

A B S T R A C T

Poultry meat is considered as a potential source of colistin resistant *Escherichia coli* (CREC). This study aimed to determine the prevalence and characteristics of CREC in broiler meat and ascertain their possible zoonotic potential(s). Broiler meat ($n = 104$) comprising 26 of each of the thigh, breast, liver, and proventriculus-gizzard was purchased from the retail outlets, Bangladesh. CREC was isolated from the meat samples on MacConkey agar plates containing colistin sulfate followed by PCR confirmation, *mcr* subtyping (*mcr-1* to *mcr-5*), phylogenetic grouping and detailed molecular characterization through whole genome sequencing (WGS). Antimicrobial resistance of the CREC isolates were evaluated by disc diffusion method and MIC (minimum inhibitory concentration) of colistin sulfate was determined by broth microdilution. The investigation revealed 58 (55.77 %) of 104 samples as positive for CREC, and 53 (91.38 %) of CREC isolates carried *mcr-1* gene with no other *mcr* subtypes evident. Most of the CREC belonged to commensal *E. coli* (66.04 %) with some pathogenic phylotypes (33.96 %) based on dichotomous decision tree. All the *mcr-1* CREC isolates were multidrug-resistant (MDR) and had MICs of 4–8 μg/mL colistin sulfate. WGS of a commensal MDR *mcr-1* CREC strain 1ChBEc2mcr revealed as a potential human pathogen belonging to ST162 that harbored 60 virulence factors associated genes (VFGs). The *mcr-1* gene in 1ChBEc2mcr genome was located on a plasmid (p1ChBEc2mcr) and showed nucleotide similarities (>95 %) to another plasmid reported from human *E. coli* in Bangladesh. Beyond *mcr-1* gene, this plasmid (p1ChBEc2mcr) also harbored genes related to aminoglycoside, beta-lactams, macrolides, and tetracycline resistance. Presence of similar *mcr-1* carrying plasmids in broiler and human CREC denotes a threat of possibly human to avian (broiler) or vice-versa transfer of *mcr-1* CREC through close contact as prevailing in the retail outlets of Bangladesh.

1. Introduction

Colistin is a reserve group of antibiotic that is used as a last resort for treating life-threatening MDR infections, particularly with the emergence of β-lactam and carbapenem resistance globally (Nation and Li, 2009). Due to widespread concerns about the emergence of resistance to this last resort, the World Health Organization (WHO) has designated it as the "Highest Priority Critically Important Antimicrobial" (World Health Organisation (WHO), 2018).

Bacteria can develop colistin resistance through an intrinsic adaptation mechanism or mutation and horizontal acquisition of the resistant genes; however, the mechanism of colistin resistance is not completely understood (Hood et al., 2014; Liu et al., 2016). Mobile colistin resistance (MCR) is critical among the resistance mechanisms as it poses additional public and environmental risks through easy and rapid spread via horizontal transfer from resistant to sensitive isolates. The emergence of mobile colistin resistance, mediated by *mcr-1*, in Gram-negative bacteria, including *E. coli* from humans and animals, was first reported in China in 2016 (Liu et al., 2016). Ten distinct colistin resistance genes (*mcr-1* to *mcr-10*) have since been discovered (Hussein et al., 2021). The transmission of *mcr* genes is facilitated by mobile genetic elements like transposons (ISApl1-PAP2-*mcr-1*-ISApl1:Tn6330) and plasmids (Liu et al., 2016; Snesrud et al., 2016). Plasmids encountered in the horizontal transfer of *mcr* genes were multireplicon types belonging to IncI2,

* Corresponding author.
E-mail address: dr_jahid@bau.edu.bd (J. Hassan).

https://doi.org/10.1016/j.ijfoodmicro.2022.110065
Received 21 October 2022; Received in revised form 15 December 2022; Accepted 23 December 2022
Available online 29 December 2022

IncP, IncX4, IncFIA, IncHI2, IncFII, IncK2 family (Donà et al., 2017). However, IncI2, IncX4, and IncHI2 plasmids majorly contributed to the transfer of *mcr-1* genes (Matamoros et al., 2017).

Colistin resistance has been attributed to the improper or excessive use of this important antibiotic in poultry production (Kempf et al., 2016). Colistin is commonly used in poultry industries since the historical era, and samples taken as far back as the 1980s have revealed the presence of bacteria resistant to colistin (Kempf et al., 2016; Shen et al., 2016). Colistin is commonly applied prophylactically through feed to promote growth in the chicken industries in Bangladesh (Islam et al., 2016), which may contribute to the emergence of colistin resistance. A series of recent studies reported a high prevalence of *mcr-1* in *E. coli* isolated from poultry fecal samples and farm environments in Bangladesh (Ahmed et al., 2020; Amin et al., 2020; Jalal et al., 2020; Sobur et al., 2019). In addition, colistin-resistant *E. coli* has been reported from urban sludge, street food, surface water, hand rinse, and human stool in Bangladesh indicating a wide range of environmental distribution of this pathogen in this country (Islam et al., 2017; Johura et al., 2020). Despite the lack of documentation about the zoonotic association of colistin-resistant *E. coli*, the presence of this MDR pathogen in a variety of environmental factors raises concerns for the public health in this country.

Colistin-resistant *E. coli* isolated from different sources in Bangladesh revealed a heterogeneous genetic pattern, but they were found to carry identical sequences of *mcr*-1 genes (Johura et al., 2020). A recent genome-based study in Bangladesh reported the isolation of *mcr-1* carrying commensal *E. coli* from the rectal swab of a patient who carried the *mcr-1* gene on a horizontally transferable plasmid (Farzana et al., 2019). These findings initiated one important question whether similar colistin-resistant determinants like plasmid or transposons are circulating among the colistin-resistant *E. coli* isolated from different sources. To answer the question, a comprehensive study with isolates from different sources and genome-based approaches are necessary. The genomic evidence along with the importance of poultry as a source of colistin-resistant *E. coli* prompted us to carry out the present study. In this study, we isolated CREC from chicken meat, elucidated the *mcr* subtyping, phylogrouping, and antibiotic resistance pattern of the isolates. In addition, WGS and downstream bioinformatics analysis was performed for characterization of a *mcr-1* CREC and to elucidate a probable connection with CREC previously reported from humans.

2. Material and methods

2.1. Sample collection

A total of 104 samples comprising 26 of each of the thigh, breast, liver, proventriculus-gizzard were collected from different retail outlets of Mymensingh City Corporation, Bangladesh from October 2021 to June 2022. Details on the samples examined in this study are furnished in Table S1.

2.2. Sample enrichment and isolation of *E. coli*

Twenty-five (25) g of meat samples were homogenized with 225 mL buffered peptone water (BPW) (Himedia, India). From the homogenate, 100 μL was spread onto MacConkey agar medium containing colistin sulfate 1 μg/mL (Fujifilm Wako Chemical Corporation, Osaka, Japan) followed by overnight (~16 h) incubation at 37 °C. After incubation, at least three (3) *E. coli* specific lactose-fermenting colonies from each plate were purified by subsequent streaking onto MacConkey agar containing colistin sulfate (1 μg/mL). Crude genomic DNA was extracted from the *E. coli* suspected isolates by boiling method followed by PCR targeting the *malB* gene using the primers furnished in Table S2 (Wang et al., 1996).

2.3. Detection and subtyping of mobile colistin resistance (mcr) genes

Multiplex PCR was used to analyze the *mcr* genes (*mcr-1* to *mcr-5*) of all the *E. coli* isolates with MICs ≥2 μg/ mL colistin sulfate (Rebelo et al., 2018). The PCR reaction was optimized to 20 μL using one Taq® Quick-Load® 2X master mix (New England Biolabs) and 5 pmol of each primer (Table S2). PCR primers and conditions employed in this study are described in Table S2.

2.4. Phylogenetic grouping of *E. coli*

A total of 53 *mcr-1* CREC, considering at least one isolate from each *mcr-1* positive sample, were grouped phylogenetically by PCR targeting *chuA*, *yjaA*, and TspE4.C2 DNA fragments (Clermont et al., 2000). PCR was conducted in a 20 μL volume with One Taq® Quick-Load® 2X master mix (Biolabs) and 10 pmol of each primer following the conditions mentioned in Table S2.

2.5. Antibiotic susceptibility testing

The antibiotic susceptibility of the isolated CREC was determined by disc diffusion method (CLSI, 2020). A total of 21 antibiotics of 11 classes frequently used in poultry farms and human clinical cases in Bangladesh, were chosen (Table S3). The experiment was performed at least three times to confirm the reproducibility of the results and *E. coli* strain ATCC25922 was used as the control strain. Isolates showing resistance to three or more classes of antibiotics are considered as MDR (Magiorakos et al., 2012).

2.6. Determination of minimum inhibitory concentration (MIC)

The MIC of colistin sulfate was determined by broth microdilution method using cation adjusted Mueller Hinton broth (Mueller Hinton broth No. 2 adjusted cation, Himedia, India) (CLSI, 2020). Test plates consisted of 100 μL of 2-fold serially diluted colistin sulfate/well (0.5–128 μg/mL) and 100 μL of respective bacterial suspension (5×10^5 CFU). After incubating at 37 °C for 20 h results were interpreted according to CLSI guidelines. MIC breakpoint of each sample against colistin sulfate was recorded where growth was significantly reduced, ignoring tiny buttons or light or faint turbidity (CLSI, 2020).

2.7. Whole genome sequencing (WGS)

WGS of a MDR *mcr-1 E. coli* strain 1ChBEc2mcr belonged to phylogroup B1 was performed on an Illumina Nextseq 550 platform (Illumina, CA, USA). Trimmomatic (Galaxy Version 0.38) was used to check the quality of FASTQ reads and trimming of low-quality residues. Trimmed reads were de novo assembled using a hybrid assembler Unicycler (Galaxy Version 0.4.8.0). The genome sequence was automatically annotated by the NCBI Prokaryotic Genomes Annotation Pipeline (PGAP). The annotated WGS data were used for sequence typing, antibiotic resistance genes (ARGs) prediction, VFGs profiling, plasmids identification, and metabolic functional analysis. PROKKA annotation (Galaxy Version 1.14.6) was used to predict the function and identification of assembled sequences against nucleotide and protein sequence database. Benchmarking Universal Single-Copy Orthologs (BUSCO, Galaxy Version 5.3.2) with 'bacteria_odb10' data set was used to measure the completeness of the genome.

The circular visualization of the *E. coli* 1ChBEc2mcr genome was performed using CGViewer (http://stothard.afns.ualberta.ca/cgview_server/). The genetic relatedness of the predicted sequence types, core genome MLST analysis (cgMLST) and bacterial source tracking of the genomes were performed using BacWGSTdb 2.0 server (http://bacdb.cn/BacWGSTdb). The Comprehensive Antibiotic Resistance Database (CARD) (https://card.mcmaster.ca/), and VirulenceFinder (http://www.mgc.ac.cn/VFs/) databases were used to detect ARGs and VFGs, respectively. The *E. coli* 1ChBEc2mcr genome was annotated against the PlasmidFinder (https://cge.food.dtu.dk/services/PlasmidFinder/) to detect plasmid(s). Geneious prime platform version 2022.1.1 (http://www.geneious.com) was used to extract the plasmid sequences carrying *mcr-1* gene in the strain 1ChBEc2mcr through reference mapping. In addition, RAST (Rapid Annotation using Subsystem Technology) server (https://rast.nmpdr.org/rast.cgi) was used to detect the genomic functional features.

3. Results

3.1. Prevalence of colistin resistant mcr-1 E. coli (mcr-1 CREC)

Enrichment followed by PCR targeting the *malB* gene revealed 58 (55.77 %) out of 104 samples positive for CREC. Out of these 58 CREC isolates, 53 (91.38 %) carried *mcr-1* gene as revealed by PCR and subsequent sequence analysis (Table 1, Figs. 1 and S1). The prevalence of CREC was significantly higher in liver and PG; however, prevalence of *mcr-1* CREC did not differ significantly based on the sample types examined (Table 1).

3.2. Phylogenetic grouping of mcr-1 CREC

Phylogenetic grouping revealed most of the study isolates as commensal *E. coli* (66.04 %) belonged to phylogenetic groups A (43.40 %) and B1 (22.64 %). Pathogenic *E. coli* belonging to the group B2 and D were evident in 13.21 % and 20.75 % cases, respectively (Table S4).

3.3. Antibiotic susceptibility

Antibiotic susceptibility testing revealed all the *mcr-1* CREC as MDR (Fig. 2). Among the MDR isolates, 92.45 % were resistant to ≥10 antibiotics of 5 to 9 different classes (Fig. 2). Broth microdilution revealed that all the *mcr-1* CREC isolates had an MIC of 4–8 μg/mL colistin sulfate.

3.4. Genomic properties of the E. coli strain 1ChBEc2mcr

The isolate was identified by KmerFinder 3.1 as *E. coli* (strain 1ChBEc2mcr), and PathogenFinder 1.1 validated its pathogenicity (0.913 out of 1.00, near to the selected value indicating higher pathogenicity). The de-novo genome assembly of *E. coli* 1ChBEc2mcr genome revealed 5,241,557 bp (base pairs) with 45× genome coverage. In the complete genome of *E. coli* 1ChBEc2mcr, 5,129,785 reads were allocated to the draft genome, giving a G+C content of 50.4 %. There were 125 contigs in the assembled genome (Accesions: Bioproject PRJNA836579 and GenBank JANATD000000000), with a N50 of 166,358 bp, L50 as 18, and six predicted plasmids. Notably, preliminary sequence analysis identified two CRISPR arrays, five prophage-related sequences, 28 predicted genomic islands (GIs), and 44 insertion sequences (ISs) (Table S5). A circular genome map (Fig. S2) from the PROKKA annotated genome using PATRIC revealed a number of gene features, including 5093 coding sequences (CDSs), 4911 protein-coding genes, 88 RNA genes, and 166 pseudogenes (Table S5). Analysis of the genome for completeness using BUSCO, indicated that the genome assembly was 100 % complete.

Table 1
Prevalence of CREC in broiler meat examined in this study.

Sample type (no.)	No. of samples positive		*P* value (Pearson Chi-Square)	
	CREC[b] (%)	*mcr-1 E. coli* (%)	CREC	*mcr-1 E. coli*
Breast (26)	10 (38.46)	10 (38.46)		
Thigh (26)	12 (46.15)	12 (46.15)		
Liver (26)	18 (69.23)	16 (61.54)	0.047[c]	0.321
[a]PG (26)	18 (69.23)	15 (57.69)		
Total (104)	58 (55.77)	53 (50.96)		

[a] PG, Proventriculus-Gizzard.
[b] CREC, Colistin Resistant *E. coli*.
[c] A p value ≤0.05 is considered as statistically significant.

Achtman's *E. coli* MLST scheme assigned the strain 1ChBEc2mcr to ST162. The phylogenetic relationships between 1ChBEc2mcr and 177 close neighbor *E. coli* strains currently available in the NCBI GenBank database, which differed by 973 alleles, were analyzed to determine the genomic epidemiological characteristics of 1ChBEc2mcr strain in a global context (Fig. 3). The cgMLST analysis showed that the MDR *E. coli* strain HH13CH (B1 phylotype) (Accession: GCF_009820925.1) belonged to ST162 and originated from chicken feces in Bangladesh (https://www.ncbi.nlm.nih.gov/assembly/GCF_009820925.1/) is the closest ancestor of the strain 1ChBEc2mcr and differ by only 192 alleles (Fig. 3, Supplementary material-2). According to epidemiological source tracking, strain 1ChBEc2mcr shares a close evolutionary relationship with many other *E. coli* strains originated from a variety of clinical samples, including those from humans and animals (e.g., cow, pig, and laying hen) with enteritis, bloody diarrhea, cystic fibrosis, metritis, and urinary tract infections. Thus, it is plausible that strain 1ChBEc2mcr may pose a zoonotic concern by infecting human or animal hosts with illness or by horizontal transfer of plasmid-linked ARGs and/or VFGs to commensal strains.

Six anticipated plasmid replicons with 98.39, 95.79, 100, 99.52, 100, and 98.64 sequence similarity to the IncFIB (AP001918), IncFIC(FII), IncHI2, IncHI2A, IncQ1, and p011 plasmids of Gram-negative bacteria were found in the *E. coli* 1ChBEc2mcr genome (Fig. S3). A plasmid sequence that was highly similar to the sequence and genetic arrangement of the *mcr-1* carrying plasmid pRS571-MCR-1.1 (Accession CP034390.1) previously published in commensal *E. coli* from human in Bangladesh was retrieved after mapping to distinct plasmids reported to carry the *mcr-1* gene (Figs. 4–5).

There were 56 ARGs in the 1ChBEc2mcr genome, which conferred resistance to colistin (*mcr-1*), beta-lactamase (*ampC1*, *ampH*), fluoroquinolones (*parC*, *gyrA*), fosfomycin (*glpT*), tetracycline (*tetA*), sulphonamide (*sul1*), acriflavine (*acrA*, *acrB*, *acrD*) and aminoglycosides (*cpxA*) (Fig. 6). In the genome of 1ChBEc2mcr, the efflux pump, which confers resistance to numerous antibiotics (MDR), was discovered to be the predominant resistance class. For instance, multidrug efflux pumps (*emrA*, *emrB*, *emrE*, *emrK*, emrY), tripartite multidrug transporter (TolC), multiple antibiotic resistance regulator (*marR*), and multidrug efflux pumps regulator (CRP) are the main mechanisms giving resistance to strain 1ChBEc2mcr (Fig. 6). Furthermore, genes encoding an aminoglycoside-modifying enzyme (*aph(3′)-Ia*, *aph(3″)-Ib*, *aph(6)-Id*, *aph(4)-Ia*, *acc(3)-Ib*) and ABC transporter ATP-binding/permease protein (YojI) were also found in the study genome (Fig. 6). Interestingly, the *mcr-1*, *aph(3′)-Ia*, *aph(3″)-Ib*, *aph(6)-I*, *aadA2*, *blaTEM*, *mph(A)*, *tetA* and *tetR* genes were located on the plasmid recovered by reference mapping to plasmid pRS571-MCR-1.1 (Fig. 4).

The genome of 1ChBEc2mcr harbored 60 VFGs with >90.0 % nucleotide similarity. Of the detected VFGs, genes associated with ferrienterobactin ABC transporters, iron-regulated enterobactin exporter, ferric enterobactin esterase, siderophore biosynthesis, *E. coli* pilus chaperone, *E. coli* pilus regulator, general secretion pathways, Type III secretion, pilus biogenesis, enterobactin synthesis were detected in the 1ChBEc2mcr genome (Table S6).

The 1ChBEc2mcr genome was annotated using the RAST server, and a total of 1569 subsystems with 30 % subsystem coverage were produced. The biggest number of genes are involved in the metabolism of carbohydrates (367 genes), followed by the metabolism of amino acids and their derivatives (321 genes), the metabolism of proteins (193 genes), and the metabolism of cofactors, vitamins, prosthetic group, and pigments (161 genes). In the subsystem analysis, the 1ChBEc2mcr genome encodes for 49 genes for virulence, disease, and defense, of which 26 (53 %) genes were linked to drug and toxin resistance, 16 (32.6 %) genes were linked to invasion and intracellular resistance, and

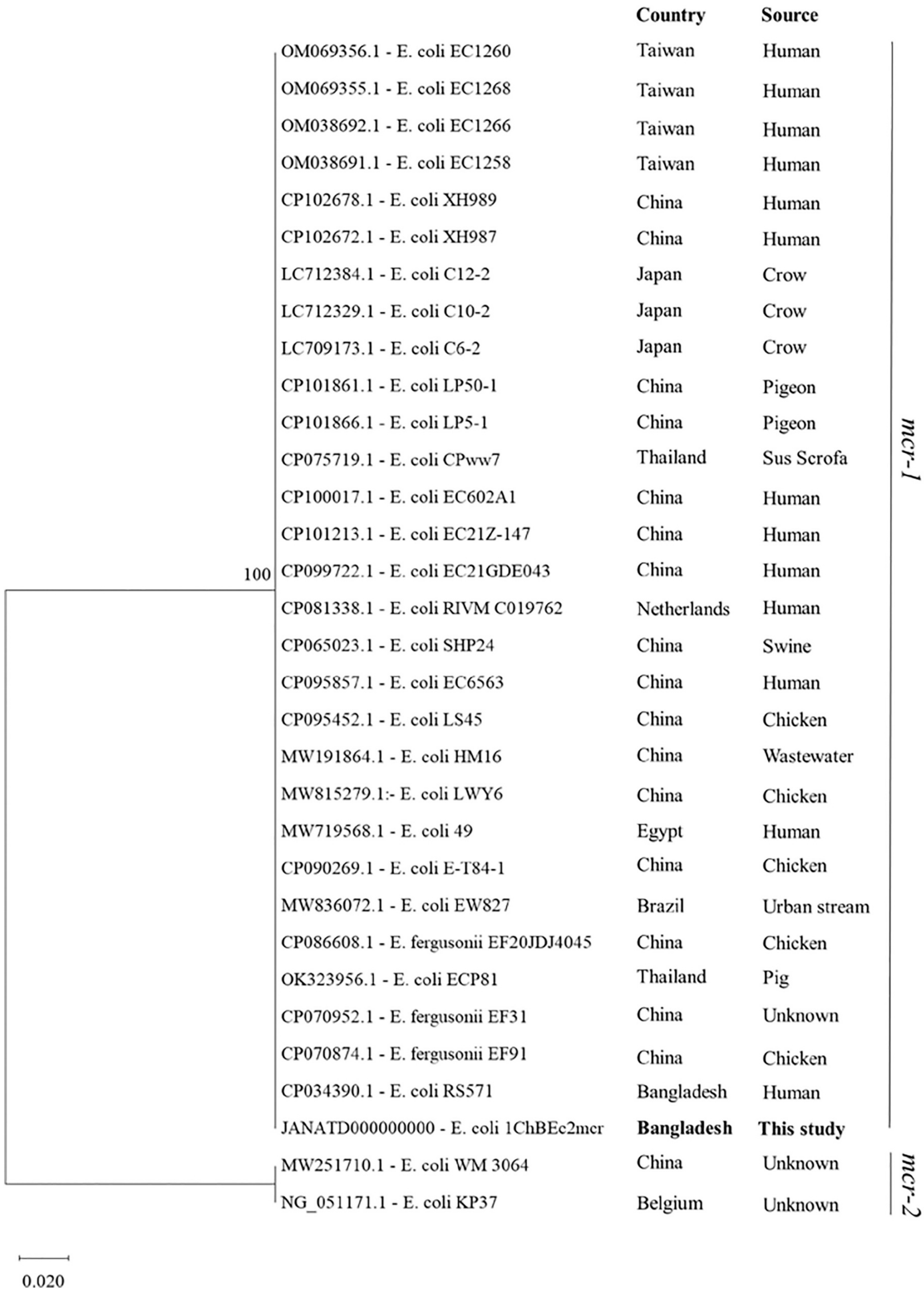


Fig. 1. Evolutionary relationship tree of the *mcr* gene sequences revealed in this study. The evolutionary relation was inferred using the Neighbor-Joining method (Saitou and Nei, 1987). The optimal tree is shown. The percentage of replicate trees in which the associated taxa clustered together in the bootstrap test (1000 replicates) is shown next to the branches (Felsenstein, 1985). The tree is drawn to scale, with branch lengths in the same units as those of the evolutionary distances used to infer the phylogenetic tree. The evolutionary distances were computed using the p-distance method (Nei and Kumar, 2000) and are in the units of the number of base differences per site. This analysis involved 32 nucleotide sequences. All ambiguous positions were removed for each sequence pair (pairwise deletion option). There were a total of 1627 positions in the final dataset. Evolutionary analyses were conducted in MEGA X (Kumar et al., 2018). The sequence whose sources are not clearly mentioned in the GenBank flat file is designated as unknown.

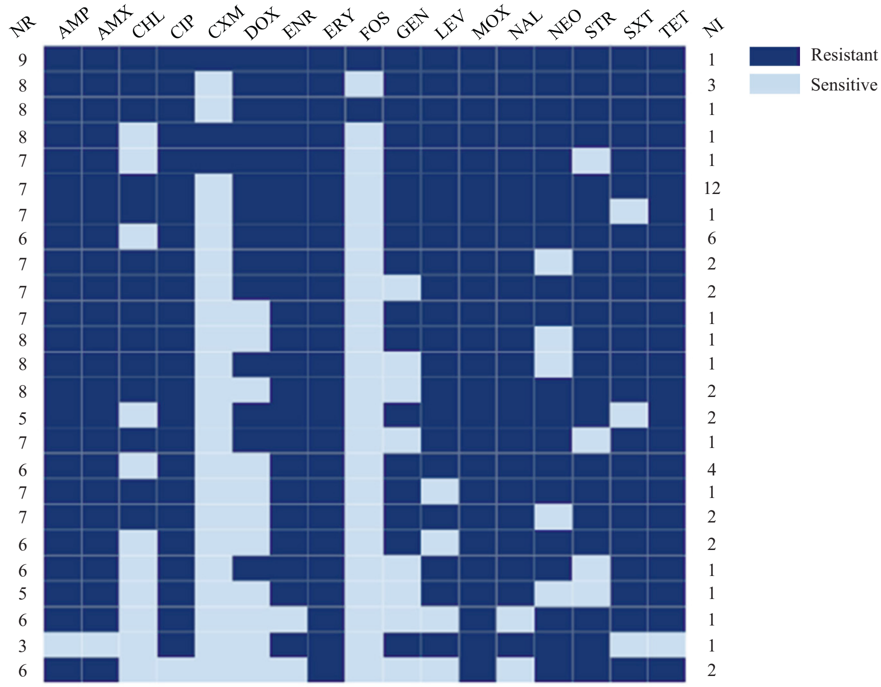


Fig. 2. Heat map showing the antimicrobial susceptibility pattern of the *mcr-1 E. coli* isolated in this study. The heat map was generated on DISPLAYR (https://southeastasia.displayr.com/). Legends: NR, Number of antibiotic resistant classes according to CLSI (CLSI, 2020); NI, Number of isolate(s); AMP, Ampicillin; AMX, Amoxicillin; CXM, Cefuroxime; CHL, Chloramphenicol; CIP, Ciprofloxacin; CRO, Ceftriaxone; DOX, Doxycycline; ENR, Enrofloxacin; ERY, Erythromycin; FOS, Fosfomycin; GEN, Gentamicin; LEV, Levofloxacin; MOX, Moxifloxacin; NAL, Nalidixic acid; NEO, Neomycin; NIT, Nitrofurantoin; STR, Streptomycin; SXT, Sulfamethoxazole-trimethoprim; TET, Tetracycline.

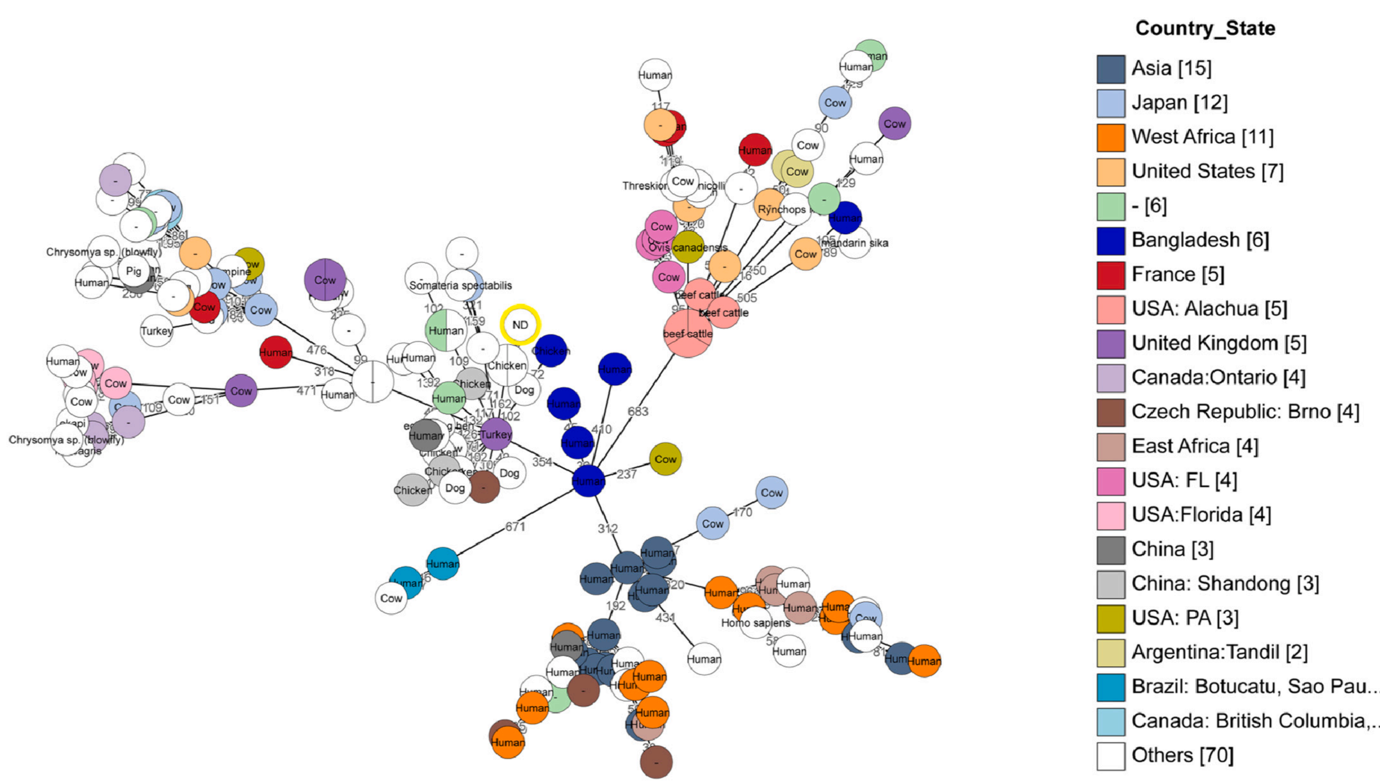


Fig. 3. Phylogenetic relationship between *E. coli* 1ChBEc2mcr and 177 close neighbors revealed by source tracking on BacWGSTdb 2.0 (http://bacdb.cn/BacWGSTdb) using cgMLST scheme, and 1000 SNP and MLST thresholds. Isolates from Bangladesh are colored in blue and the isolate 1ChBEc2mcr is highlighted in bright yellow color. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2 (4 %) genes were linked to bacteriocins, ribosomally synthesized antibacterial peptides (Fig. S4).

4. Discussion

Colistin resistant *E. coli* (CREC) is an emerging public health concern worldwide (Zhang et al., 2021). CREC existed in a variety of environmental setting including poultry habitats and live bird markets who act

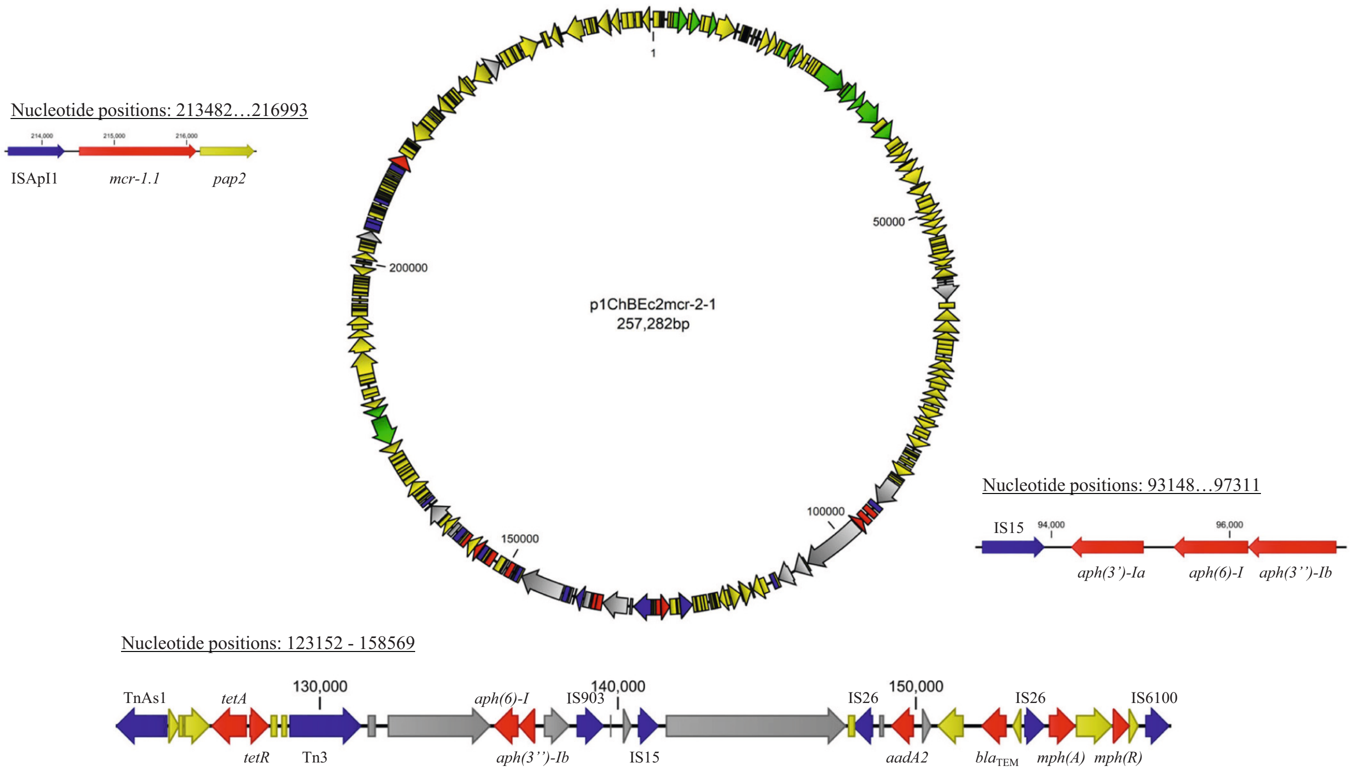


Fig. 4. Structural organization of ORFs in the plasmid p1ChBEc2mcr. The plasmid was generated by mapping raw read of strain 1ChBEc2mcr with plasmid pRS571-MCR-1.1 (Accession no. CP034390) on Geneious Prime platform (Version 2022.1.1). Genes coding for antibiotic resistance are colored in red, transposons in blue, conjugation or transfer in green and others in yellow colors. Ash color indicates assembly gaps resulted during reference mapping with plasmid pRS571-MCR-1.1. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

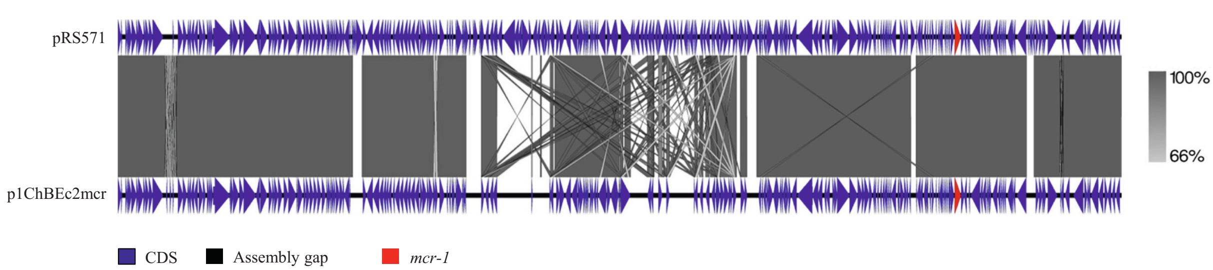


Fig. 5. Linear comparison of plasmid sequence p1ChBEc2mcr with *mcr-1.1* carrying plasmid sequences reported from human stool in Bangladesh (Farzana et al., 2019). The sequence of p1ChBEc2mcr was derived by mapping the raw reads of strain 1ChBEc2mcr with pRS571-MCR-1.1 (Accession no. CP034390).

as a potential source of transmission to human (Amin et al., 2020; Trung et al., 2017). Chicken meat is the most often reported human source of CREC (Aklilu and Raman, 2020; Chiba et al., 2019; Joshi et al., 2019; Monte et al., 2017; Nakayama et al., 2022; Schrauwen et al., 2017); however, the occurrence of CREC in chicken meat was not described in Bangladesh. Therefore, we have attempted to isolate CREC from chicken meat sold at the retail outlets and elucidate the characteristics of CREC carrying *mcr* genes through phenotypic and genotypic approaches.

In this study, 55.77 % of the chicken meat was positive for CREC and 91.38 % of the CREC carried *mcr-1* gene. This study recorded a higher prevalence of *mcr-1* CREC than other publications from Brazil (19.5 %) (Monte et al., 2017), Japan (9.7 %) (Odoi et al., 2021), Netherlands (24.8 %) (Schrauwen et al., 2017), Nepal (26.6 %) (Joshi et al., 2019), and Vietnam (43.3 %) (Nakayama et al., 2022). Despite the fact that no researchers yet examined the presence of CREC in chicken meat, *mcr-1 E. coli* was found in feces collected from 13.5 % of poultry environment (Amin et al., 2020) and 25 % of broiler birds (Ahmed et al., 2020) in Bangladesh. Cross contamination due to poor hygiene at the slaughtering and dressing facilities might attribute to the higher prevalence of *mcr-1 E. coli* in chicken meat in this study (Mensah et al., 2022).

Almost all of the *mcr-1* CRECs isolated in this study was resistant to majority of the antibiotics used in Bangladesh's poultry industries, including the Highest and High Priority Critically Important Antimicrobials (HP-CIA) for humans (WHO, 2018). Similar MDR resistance was previously described in *E. coli* isolated from chicken meat (Parvin et al., 2020) and *mcr-1 E. coli* isolated from poultry environments (Amin et al., 2020). Most of the MDR *mcr-1* CREC isolated in this study was commensal (66.04 %) which is higher than those previously reported in broiler birds from Bangladesh (25 %) (Ahmed et al., 2020). Occurrence of MDR and colistin resistance genes in the commensal *E. coli* might be attributed to the imprudent use of antibiotics in the poultry farms. Studies from Bangladesh showed that 84.7 % farmers use antibiotics in the poultry farms for prophylactic purposes apart from therapeutic uses (Imam et al., 2020). In commercial poultry farming 24 different types of antimicrobials have been reported from Bangladesh where 56.6 % of the broiler farms used colistin for prophylaxis or growth promotion (Imam

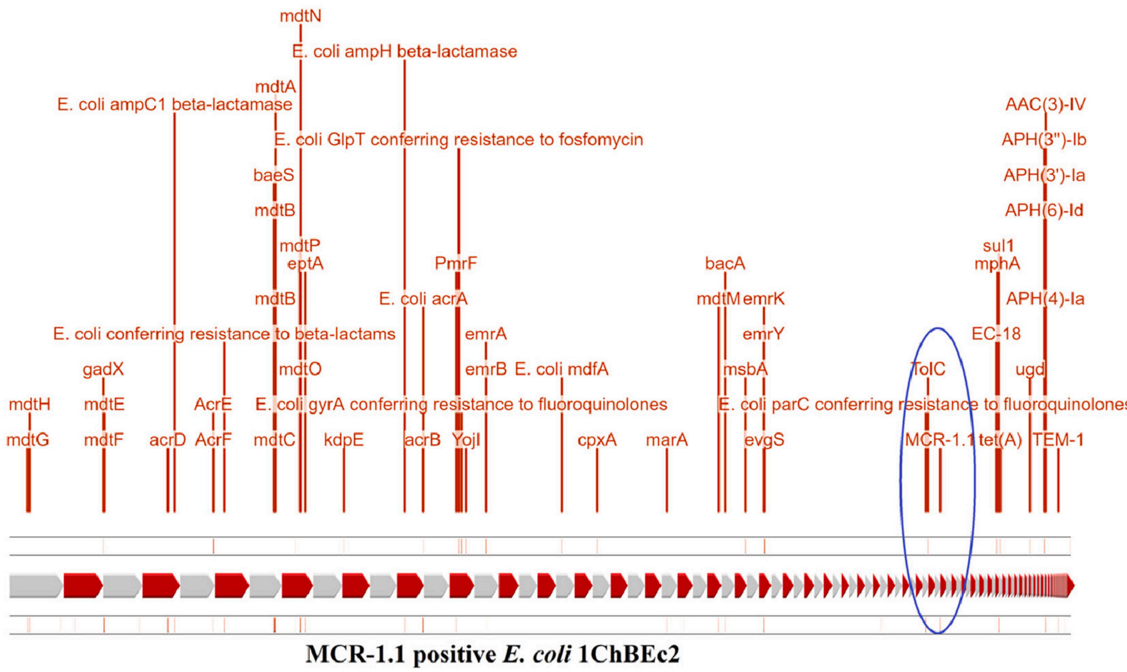


Fig. 6. Linear representation of antimicrobial resistance genes (ARGs) and related pathways in the *E. coli* 1ChBEc2mcr genome found through CARD (Comprehensive Antibiotic Resistance Database). ARGs and related pathways found in the genome are presented in red color. The outermost and innermost layers (black) show the forward and reverse strand of the genome. Relative location of the antimicrobial resistances genes and related pathways in the genome are shown in light red bar inside the black layers. Arrows (red and grey) in the middle denotes different contigs of the genome. The MCR-1.1 gene is highlighted in blue color ellipse. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

et al., 2020). Occurrence of MDR *mcr-1* CREC in chicken meat or poultry farm environment might be resulted from the selective pressure imposed by the injudicious use of antibiotics in poultry production (Islam et al., 2016; Nguyen et al., 2016).

Whole genome sequencing (WGS) and subsequent bioinformatic analysis of one MDR *mcr-1* commensal CREC (strain 1ChBEc2mcr) was performed to determine its genetic characteristics and possession of plasmid carrying *mcr-1* gene. The *E. coli* sequenced in this study belonged to ST162 which is closely related to a MDR commensal *E. coli* isolated from chicken feces in Bangladesh, (Unpublished, https://www.ncbi.nlm.nih.gov/assembly/GCF_009820925.1/), indicating the isolate is a chicken origin. Moreover, this strain has a close evolutionary relationship with *E. coli* isolated previously from human and animal disease, indicating a potential threat of zoonosis.

The genome (1ChBEc2mcr) carried *mcr-1* gene on a mega-plasmid (~257 kb) of IncHI2 family as revealed by reference mapping. Occurrence of *mcr-1* gene on IncHI2 plasmid has been reported earlier (Ahmed et al., 2020; Matamoros et al., 2017). The *mcr-1* gene was flanked by ISApI1 and *pap2* on either side. ISApI1 is known as a *mcr-1* gene mobilizing transposon component which plays a pivotal role in the spread of *mcr-1* gene (Zurfluh et al., 2017). Similar ISApI1 replicon has been reported in *mcr-1 E. coli* isolated from Bangladesh (Ahmed et al., 2020); however, occurrence of *mcr-1* in ~257 kb plasmid has not been reported from poultry sources in Bangladesh. Surprisingly, commensal *E. coli* carrying identical *mcr-1* gene on IncHI2 plasmid (pRS571-MCR-1.1) has been reported from the rectal swab of a patient who died of septicemia in Bangladesh (Farzana et al., 2019). Although some gap was evidenced in the sequence, the recovered sequences and genetic arrangement of the p1ChBEc2mcr was almost identical to that of pRS571-MCR-1.1. Occurrence of similar type of *mcr-1* carrying plasmid in human and poultry isolates indicated a possible transmission of this plasmid from poultry to human or vice-versa. However, further investigation with isolates from human and animal sources are necessary to confirm this possibility.

The plasmid p1ChBEc2mcr carries genes related to aminoglycoside resistance [*aph(3′)-Ia*, *aph(3″)-Ib*, *aph(6)-I*, *aadA2*], β-lactam resistance (bla_{TEM}), tetracycline resistance (*tetA*, *tetR*) and macrolide resistance gene *mph(A)* in addition to the *mcr-1* gene similar to that of pRS571-MCR-1.1 (Farzana et al., 2019). Detection of *mcr-1* gene in plasmids carrying multiple antibiotic resistant determinants has also been described earlier (Haenni et al., 2016; Yao et al., 2016).

Sequencing of more isolates from different phylogroup would provide more detailed information on the genomic or plasmid diversity of *mcr-1* CREC; however, this was not possible due to mere fund limitations. Despite several limitations this study for the first time described the occurrence of MDR CREC carrying *mcr-1* gene from chicken meat in Bangladesh, as well as, explored the presence of *mcr-1* gene on a plasmid which have probable link with *mcr-1.1* carrying plasmid described in commensal *E. coli* isolated from a patient died of septicemia in Bangladesh. However, to ascertain zoonotic transmission of CREC or the *mcr-1* plasmid from chicken meat to human further investigation with more samples from humans and animals covering large geographical locations is necessary.

5. Conclusions

For the first time in Bangladesh, we identified and characterized MDR colistin-resistant *E. coli* from chicken meat carrying *mcr-1* gene. The data, despite of some limitations, provides a meaningful picture of the *mcr-1* gene for colistin resistance, highlighting the problem of transferable colistin resistance and the public health risk in Bangladesh. It is necessary to closely monitor and strictly regulate the use of antimicrobials in Bangladeshi chicken production because colistin resistance in *E. coli* may be caused by the improper use of colistin sulfate in the poultry business.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijfoodmicro.2022.110065.

Funding

This work was performed with partial support from the fund provided by the Ministry of Education (MoE), Government of Bangladesh (Project No. LS20191223), and Bangladesh Agricultural University Research Systems (BAURES) (Project No. 2021/21/BAU).

Ethical approval

This study design was approved by the Animal Welfare and Experimentation Ethics Committee, Bangladesh Agricultural University [Approval no. AWEEC/BAU/2021 (48)].

Declaration of competing interest

The authors know no competing interest to declare.

Data availability

Data will be made available on request.

Acknowledgement

The authors are highly indebted to the CLC genomic workbench (Qiagen), Geneious prime (Geneious) and DNASTAR Lasegene (DNASTAR Inc., USA) team for providing free trial license for analyzing and verifying the genomic data described in this study. The authors also take the chance to thank CSO, Antimicrobial Resistance Action Centre, Bangladesh Livestock Research Institute, Savar, Dhaka, for providing *E. coli* ATCC strain used in this study.

References

Ahmed, S., Das, T., Islam, M.Z., Herrero-Fresno, A., Biswas, P.K., Olsen, J.E., 2020. High prevalence of mcr-1-encoded colistin resistance in commensal Escherichia coli from broiler chicken in Bangladesh. Sci. Rep. 10 https://doi.org/10.1038/S41598-020-75608-2.

Aklilu, E., Raman, K., 2020. MCR-1 gene encoded colistin-resistant Escherichia coli in raw chicken meat and bean sprouts in Malaysia. Int. J. Microbiol. 2020 https://doi.org/10.1155/2020/8853582.

Amin, M.B., Sraboni, A.S., Hossain, M.I., Roy, S., Mozmader, T.A.U., Unicomb, L., Rousham, E.K., Islam, M.A., 2020. Occurrence and genetic characteristics of mcr-1-positive colistin-resistant E. coli from poultry environments in Bangladesh. J. Glob. Antimicrob. Resist. 22, 546–552. https://doi.org/10.1016/J.JGAR.2020.03.028.

Chiba, N., Tanimoto, K., Hisatsune, J., Sugai, M., Shibayama, K., Watanabe, H., Tomita, H., 2019. Detection of mcr-1-mediated colistin resistance in E.coli isolate from imported chicken meat from Brazil. J. Glob. Antimicrob. Resist. 16, 249–250. https://doi.org/10.1016/J.JGAR.2019.02.004.

Clermont, O., Bonacorsi, S., Bingen, E., 2000. Rapid and simple determination of the Escherichia coli phylogenetic group. Appl. Environ. Microbiol. 66, 4555–4558. https://doi.org/10.1128/AEM.66.10.4555-4558.2000.

CLSI, 2020. M100-Ed30 - Performance Standards for Antimicrobial Susceptibility Testing - 30th Edition [WWW Document]. n.d. URL (accessed 10.3.22). https://webstore.ansi.org/Standards/CLSI/CLSIM100Ed30.

Donà, V., Bernasconi, O.J., Pires, J., Collaud, A., Overesch, G., Ramette, A., Perreten, V., Endimiani, A., 2017. Heterogeneous genetic location of mcr-1 in colistin-resistant Escherichia coli isolates from humans and retail chicken meat in Switzerland: emergence of mcr-1-carrying IncK2 plasmids. Antimicrob. Agents Chemother. 61 https://doi.org/10.1128/AAC.01245-17.

Farzana, R., Jones, L.S., Rahman, M.A., Toleman, M.A., Sands, K., Portal, E., Boostrom, I., Kalam, M.A., Hassan, B., Uddin, A.N., Walsh, T.R., 2019. Emergence of mcr-1 mediated colistin resistant Escherichia coli from a hospitalized patient in Bangladesh. J. Infect. Dev. Ctries. 13, 773–776. https://doi.org/10.3855/JIDC.11541.

Felsenstein, J., 1985. Confidence limits on phylogenies: an approach using the bootstrap. Evolution 39, 783–791.

Haenni, M., Poirel, L., Kieffer, N., Châtre, P., Saras, E., Métayer, V., Dumoulin, R., Nordmann, P., Madec, J.Y., 2016. Co-occurrence of extended spectrum β lactamase and MCR-1 encoding genes on plasmids. Lancet Infect. Dis. 16, 281–282. https://doi.org/10.1016/S1473-3099(16)00007-4.

Hood, M.I., Becker, K.W., Roux, C.M., Dunman, P.M., Skaar, E.P., 2014. Genetic determinants of intrinsic colistin tolerance in Acinetobacter baumanii. Infect. Immun. 82, 469. https://doi.org/10.1128/IAI.01359-13.

Hussein, N.H., AL-Kadmy, I.M.S., Taha, B.M., Hussein, J.D., 2021. Mobilized colistin resistance (mcr) genes from 1 to 10: a comprehensive review. Mol. Biol. Rep. 48, 2897–2907. https://doi.org/10.1007/S11033-021-06307-Y/FIGURES/7.

Imam, T., Gibson, J.S., Foysal, M., Das, S.B., Gupta, S.Das, Fournié, G., Hoque, M.A., Henning, J., 2020. A cross sectional study of antimicrobial usage on commercial broiler and layer chicken farms in Bangladesh. Front. Vet. Sci. 7 https://doi.org/10.3389/FVETS.2020.576113.

Islam, K.B.M.S., Islam, K.B.M.S., Shiraj-Um-Mahmuda, S., Hazzaz-Bin-Kabir, M., 2016. Antibiotic usage patterns in selected broiler farms of Bangladesh and their public health implications. J. Public HealthDev. Ctries. 2, 276–284.

Islam, A., Rahman, Z., Monira, S., Rahman, M.A., Camilli, A., George, C.M., Ahmed, N., Alam, M., 2017. Colistin resistant Escherichia coli carrying mcr-1 in urban sludge samples: Dhaka,Bangladesh. Gut Pathog. 9 https://doi.org/10.1186/S13099-017-0227-4.

Jalal, M.S., Dutta, A., Das, T., Islam, M.Z., 2020. First detection of plasmid-mediated colistin-resistance gene (mcr-1, mcr-2 and mcr-3) in Escherichia coli isolated from breeder poultry of Bangladesh. Int. J. Infect. Dis. 101, 17. https://doi.org/10.1016/j.ijid.2020.09.082.

Johura, F.T., Tasnim, J., Barman, I., Biswas, S.R., Jubyda, F.T., Sultana, M., George, C.M., Camilli, A., Seed, K.D., Ahmed, N., Alam, M., 2020. Colistin-resistant Escherichia coli carrying mcr-1 in food, water, hand rinse, and healthy human gut in Bangladesh. Gut Pathog. 12 https://doi.org/10.1186/S13099-020-0345-2.

Joshi, P.R., Thummeepak, R., Leungtongkam, U., Pooarlai, R., Paudel, S., Acharya, M., Dhital, S., Sitthisak, S., 2019. The emergence of colistin-resistant Escherichia coli in chicken meats in Nepal. FEMS Microbiol. Lett. 366 https://doi.org/10.1093/FEMSLE/FNZ237.

Kempf, I., Jouy, E., Chauvin, C., 2016. Colistin use and colistin resistance in bacteria from animals. Int. J. Antimicrob. Agents 48, 598–606. https://doi.org/10.1016/J.IJANTIMICAG.2016.09.016.

Kumar, S., Stecher, G., Li, M., Knyaz, C., Tamura, K., 2018. MEGA X: molecular evolutionary genetics analysis across computing platforms. Mol. Biol. Evol. 35, 1547–1549.

Liu, Y.Y., Wang, Y., Walsh, T.R., Yi, L.X., Zhang, R., Spencer, J., Doi, Y., Tian, G., Dong, B., Huang, X., Yu, L.F., Gu, D., Ren, H., Chen, X., Lv, L., He, D., Zhou, H., Liang, Z., Liu, J.H., Shen, J., 2016. Emergence of plasmid-mediated colistin resistance mechanism MCR-1 in animals and human beings in China: a microbiological and molecular biological study. Lancet Infect. Dis. 16, 161–168. https://doi.org/10.1016/S1473-3099(15)00424-7.

Magiorakos, A.P., Srinivasan, A., Carey, R.B., Carmeli, Y., Falagas, M.E., Giske, C.G., Harbarth, S., Hindler, J.F., Kahlmeter, G., Olsson-Liljequist, B., Paterson, D.L., Rice, L.B., Stelling, J., Struelens, M.J., Vatopoulos, A., Weber, J.T., Monnet, D.L., 2012. Multidrug-resistant, extensively drug-resistant and pandrug-resistant bacteria: an international expert proposal for interim standard definitions for acquired resistance. Clin. Microbiol. Infect. 18, 268–281. https://doi.org/10.1111/j.1469-0691.2011.03570.x.

Matamoros, S., Van Hattem, J.M., Arcilla, M.S., Willemse, N., Melles, D.C., Penders, J., Vinh, T.N., Thi Hoa, N., Bootsma, M.C.J., Van Genderen, P.J., Goorhuis, A., Grobusch, M., Molhoek, N., Oude Lashof, A.M.L., Stobberingh, E.E., Verbrugh, H.A., De Jong, M.D., Schultsz, C., 2017. Global phylogenetic analysis of Escherichia coli and plasmids carrying the mcr-1 gene indicates bacterial diversity but plasmid restriction. Sci. Rep. 7 https://doi.org/10.1038/S41598-017-15539-7.

Mensah, G.I., Adjei, V.Y., Vicar, E.K., Atsu, P.S., Blavo, D.L., Johnson, S.A.M., Addo, K.K., 2022. Safety of retailed poultry: analysis of antibiotic resistance in Escherichia coli from raw chicken and poultry fecal matter from selected farms and retail outlets in Accra,Ghana. Microbiol. Insights 15, 117863612210932. https://doi.org/10.1177/11786361221093278.

Monte, D.F., Mem, A., Fernandes, M.R., Cerdeira, L., Esposito, F., Galvão, J.A., Franco, B D.G.M., Lincopan, N., Landgraf, M., 2017. Chicken meat as a reservoir of colistin resistant Escherichia coli carrying mcr-1 genes in South America. Antimicrob. Agents Chemother. 61 https://doi.org/10.1128/AAC.02718-16.

Nakayama, T., Le Thi, H., Thanh, P.N., Minh, D.T.N., Hoang, O.N., Hoai, P.H., Yamaguchi, T., Jinnai, M., Do, P.N., Van, C.D., Kumeda, Y., Hase, A., 2022. Abundance of colistin-resistant Escherichia coli harbouring mcr-1 and extended-spectrum β-lactamase-producing E. Coli co-harbouring blaCTX-M-55 or -65 with blaTEM isolates from chicken meat in Vietnam. Arch. Microbiol. 204 https://doi.org/10.1007/S00203-021-02746-0.

Nation, R.L., Li, J., 2009. Colistin in the 21st century. Curr. Opin. Infect. Dis. 22, 535. https://doi.org/10.1097/QCO.0B013E328332E672.

Nei, M., Kumar, S., 2000. Molecular Evolution and Phylogenetics. Oxford University Press, New York.

Nguyen, N.T., Nguyen, H.M., Nguyen, C.V., Nguyen, T.V., Nguyen, M.T., Thai, H.Q., Ho, M.H., Thwaites, G., Ngo, H.T., Baker, S., Carrique-Mas, J., 2016. Use of colistin and other critical antimicrobials on pig and chicken farms in Southern Vietnam and its association with resistance in commensal Escherichia coli. Appl. Environ. Microbiol. 82, 3727–3735. https://doi.org/10.1128/AEM.00337-16.

Odoi, J.O., Takayanagi, S., Sugiyama, M., Usui, M., Tamura, Y., Asai, T., 2021. Prevalence of colistin resistant bacteria among retail meats in Japan. Food Saf. (Tokyo, Japan) 9, 48–56. https://doi.org/10.14252/FOODSAFETYFSCJ.D-21-00002.

Parvin, M.S., Talukder, S., Ali, M.Y., Chowdhury, E.H., Rahman, M.T., Islam, M.T., 2020. Antimicrobial resistance pattern of Escherichia coli isolated from frozen chicken meat in Bangladesh. Pathog. (Basel, Switzerland) 9. https://doi.org/10.3390/PATHOGENS9060420.

Rebelo, A.R., Bortolaia, V., Kjeldgaard, J.S., Pedersen, S.K., Leekitcharoenphon, P., Hansen, I.M., Guerra, B., Malorny, B., Borowiak, M., Hammerl, J.A., Battisti, A., Franco, A., Alba, P., Perrin-Guyomard, A., Granier, S.A., de Frutos, C., Escobar, Malhotra-Kumar, S., Villa, L., Carattoli, A., Hendriksen, R.S., 2018. Multiplex PCR for detection of plasmid-mediated colistin resistance determinants, mcr-1, mcr-2,

mcr-3, mcr-4 and mcr-5 for surveillance purposes. Euro Surveill. 23. https://doi.org/10.2807/1560-7917.ES.2018.23.6.17-00672.
Saitou, N., Nei, M., 1987. The neighbor-joining method: a new method for reconstructing phylogenetic trees. Mol. Biol. Evol. 4, 406–425.
Schrauwen, E.J.A., Huizinga, P., van Spreuwel, N., Verhulst, C., Kluytmans-van den Bergh, M.F.Q., Kluytmans, J.A.J.W., 2017. High prevalence of the mcr-1 gene in retail chicken meat in the Netherlands in 2015. Antimicrob. Resist. Infect. Control 6, 1–5. https://doi.org/10.1186/S13756-017-0242-8/TABLES/2.
Shen, Z., Wang, Y., Shen, Y., Shen, J., Wu, C., 2016. Early emergence of mcr-1 in Escherichia coli from food-producing animals. Lancet Infect. Dis. 16, 293. https://doi.org/10.1016/S1473-3099(16)00061-X.
Snesrud, E., He, S., Chandler, M., Dekker, J.P., Hickman, A.B., McGann, P., Dyda, F., 2016. A model for transposition of the colistin resistance gene mcr-1 by ISApl1. Antimicrob. Agents Chemother. 60, 6973–6976. https://doi.org/10.1128/AAC.01457-16.
Sobur, M.A., Ievy, S., Haque, Z.F., Nahar, A., Zaman, S.B., Rahman, M.T., 2019. Emergence of colistin-resistant Escherichia coli in poultry, house flies, and pond water in Mymensingh,Bangladesh. J. Adv. Vet. Anim. Res. 6, 50–53. https://doi.org/10.5455/JAVAR.2019.F311.
Trung, N.V., Matamoros, S., Carrique-Mas, J.J., Nghia, N.H., Nhung, N.T., Chieu, T.T.B., Mai, H.H., van Rooijen, W., Campbell, J., Wagenaar, J.A., Hardon, A., Mai, N.T.N., Hieu, T.Q., Thwaites, G., de Jong, M.D., Schultsz, C., Hoa, N.T., 2017. Zoonotic transmission of mcr-1 colistin resistance gene from small-scale poultry farms, Vietnam. Emerg. Infect. Dis. 23, 529. https://doi.org/10.3201/EID2303.161553.
Wang, R.F., Cao, W.W., Cerniglia, C.E., 1996. PCR detection and quantitation of predominant anaerobic bacteria in human and animal fecal samples. Appl. Environ. Microbiol. 62, 1242–1247. https://doi.org/10.1128/AEM.62.4.1242-1247.1996.
World Health Organisation (WHO), 2018. Critically Important Antimicrobials for Human Medicine 6th Re.vision 2018. Ranking of medically important antimicrobials for risk management of antimicrobial resistance due to non-human use, ISBN 978-92-4-151552-8. https://apps.who.int/iris/bitstream/handle/10665/312266/9789241515528-eng.pdf?ua=1.
Yao, X., Doi, Y., Zeng, L., Lv, L., Liu, J.H., 2016. Carbapenem-resistant and colistin-resistant Escherichia coli co-producing NDM-9 and MCR-1. Lancet Infect. Dis. 16, 288–289. https://doi.org/10.1016/S1473-3099(16)00057-8.
Zhang, S., Abbas, M., Rehman, M.U., Wang, M., Jia, R., Chen, S., Liu, M., Zhu, D., Zhao, X., Gao, Q., Tian, B., Cheng, A., 2021. Updates on the global dissemination of colistin-resistant Escherichia coli: an emerging threat to public health. Sci. Total Environ. 799 https://doi.org/10.1016/J.SCITOTENV.2021.149280.
Zurfluh, K., Nüesch-Inderbinen, M., Klumpp, J., Poirel, L., Nordmann, P., Stephan, R., 2017. Key features of mcr-1-bearing plasmids from Escherichia coli isolated from humans and food. Antimicrob. Resist. Infect. Control 6. https://doi.org/10.1186/S13756-017-0250-8.